

AUTONOMOUS KINETIC DEFENSE PLATFORM

R.A.S.K. 01 // RAVEN SERIES

Field Operator Manual & Engineering Blueprint Summary

“Efficiency in the field is a matter of design.”

System Overview

The **R.A.S.K. 01** (Rapid Armor Security and Kinematics) is an autonomous, heavy-duty kinetic security platform designed for hazardous environments, civil pacification, and active search-and-rescue. Built on a 4-legged articulated base, it provides steady elevation over irregular debris, structural ruins, and urban stairs.

Key Functional Highlights

- **Kinematic Locomotion:** Quadrupedal multi-axis legs with dynamic gyro-stabilization.
- **Situational Awareness:** 360-degree spherical dome with dual LiDAR, thermal FLIR, and acoustic mapping.
- **Modular Payload Arm:** Integrated non-lethal deterrents (taser, strobe, aerosol) and optional kinetic module.
- **Armor Matrix:** Certified Level IV composite plates with deployable high-absorption riot shield.
- **Rear Extraction Unit:** Sealed life-support rescue capsule for casualty evacuation.

Dimensional & Operational Specifications

Parameter	Specification	Operational Context
Total Height	1120 mm	Low-profile center of mass for anti-rollover
Shoulder Width	820 mm	Compatible with standard commercial door frames
Chassis Depth	760 mm	Balanced footprint for stair climbing
Dry Mass	180 kg	High inertial resistance against physical concussions
Chassis Material	Al-Alloy + Hardened Steel	Multi-layer ballistic and corrosion resistance
Degrees of Freedom	16 (8 per arm, 4 per leg)	High mechanical manipulation agility
Power Plant	Li-Ion 48V / 10 kWh	Hermetically sealed interior battery bank
Operational Autonomy	Up to 8 hours	Continuous perimeter monitoring profile
Maximum Velocity	6 km/h	Controlled tactical pacing alongside human officers
Payload Capacity	50 kg	External casualty drag or supply conveyance

Schematic Overview & Blueprints

Below are the primary orthogonal projections extracted from the assembly master sheet (Rev. A).

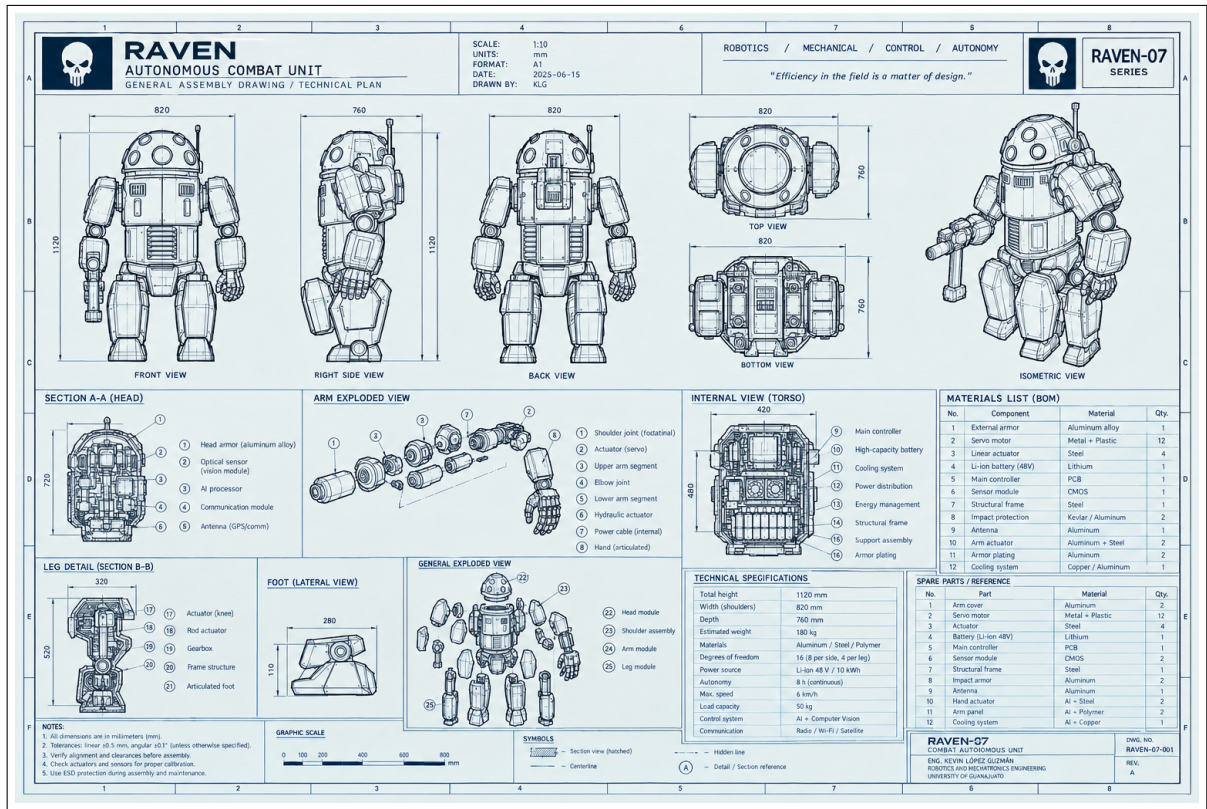


Figure 1: General Assembly Blueprint (M-07 Architecture).

Bill of Materials Summary

Item No.	Sub-Assembly	Material / Standard	Quantity
01	External Armor Shell	7075-T6 Aluminum Alloy	1 Unit
02	High-Torque Servomotors	High-Density Coreless / Steel Gears	12 Units
03	Linear Leg Actuators	Hardened Carbon Steel	4 Units
04	Energy Storage Bank	Lithium-Ion 48V 10kWh	1 Pack
05	Main ECU & AI Controller	Triple-redundant embedded PCB	1 Unit
06	Optical & LiDAR Array	CMOS Sensor + Solid-State LiDAR	2 Clusters
07	Internal Frame	High-Tensile Structural Steel	1 Core
08	Articulated Hand Limbs	Aluminum + Synthetic High-Grip Rubber	2 Assemblies
09	Sealed Cooling System	Dual Copper Heatpipe + Active Blower	1 Unit

Safety Protocols for Field Personnel

- Safe Approach Distance:** Maintain at least 1.5 meters from articulated knee joints while dynamic stabilization mode is engaged.
- Emergency Neutralization:** A mechanical disconnect switch is located behind the upper dorsal thermal exhaust port.
- Lethal Force Protocol:** Force-escalation options require 2-step cryptographically authenticated manual confirmation from the remote human operator.